

Data Handling: Import, Cleaning and Visualisation

Exercise 5: Data Reshaping, Cleaning, and Merging

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Exercise A: Reshaping Data

In this exercise, we will learn how to reshape data into long format using the built-in `economics` dataset from the `ggplot2` package. Load the dataset into your R session. It contains monthly data on several indicators of the U.S. economy:

Variable	Description
<code>date</code>	month of data collection
<code>pce</code>	personal consumption expenditures (billions of USD)
<code>pop</code>	total population (thousands)
<code>psavert</code>	personal savings rate
<code>unempmed</code>	median duration of unemployment (weeks)
<code>unemploy</code>	number of unemployed (thousands)

A.1 - Convert to Long Format

Reshape the dataset into long format, so that for each date, there are multiple rows — one for each of the other variables. Save the result as `df_long`.

A.2 - Convert to Wide Format

From the new `df_long` dataset, reshape the data back to its original wide format, and save it as `df`.

Exercise B: Cleaning Data

In this section, we'll practice three important data-cleaning functions from `dplyr`: `filter()`, `select()`, and `mutate()`.

B.1 - Filter Observations

Start with data `df`. We are only interested in observations collected after the year 2000. Filter the data accordingly and save the result as `df_reduced`.

B.2 - Select Observations

Now we are only interested in the evolution of `pce` and `pop` over time. Clean the data accordingly and save the new dataset `df_reduced`.

B.3 - Create Indicator Variable

We want to add a variable that marks all observations collected after the year 2000. Create a new variable called `after_2000` and add it to `df_reduced`.

Exercise C: Merging Data

In the final exercise, we will learn how to merge two dataframes using `left_join()` from `dplyr`.

C.1 - Merge Datasets

We want to flag the original `economics` dataset with a new variable indicating whether each observation was collected after 2000. Use `left_join()` to merge `df` (the original dataset) with `df_reduced` (which contains `after_2000`). Which dataset should be used as the master (left) table, and which one as the joining (right) table?

C.2 - Handle Missing Values

The merge created some missing values for the variable `after_2000`. Explain why this could happen, and propose a solution to have a tidy dataset.